

# Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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## Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

**Keywords:** Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

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in exposed occupations. While their analysis covers 25 broad occupation groups, we exploit variation across a much wider variety of job types, as well as a firm-level analysis that examines within-firm dynamics that would be invisible in occupational aggregates. In addition, our individual-level posting data enables examination of a wider set of outcome variables such as salary distributions and seniority-specific employment, as well as heterogeneity in treatment effects by pre-treatment compensation or location.

Taken together, our findings raise important policy questions. The disproportionate elimination of entry-level positions disrupts the conventional model of skill development, where employees master progressively complex tasks through hands-on experience with routine work. Without junior roles that serve as training grounds, firms may struggle to develop future senior talent internally, creating a hollowed-out labor market where entry becomes increasingly difficult (Ide, 2025). This dynamic could exacerbate inequality as those with pre-existing experience benefit from reduced competition while new entrants face shrinking opportunities. Policy responses must address both immediate displacement and longer-term skill development pathways.

## 2 Data

Our primary data comes from Revelio Labs, which aggregates and processes publicly available employment information from professional networking platforms, company websites, and job boards using machine learning algorithms to create proprietary longitudinal labor market databases. Revelio Labs transforms semi-structured public profile information into a structured panel dataset through systematic parsing, standardization, and cleaning procedures. Their data have been validated against official statistics and used in recent academic research on labor market dynamics (Fadhel et al., 2021; Eisfeldt et al., 2023; Hampole et al., 2025). In Appendix A4, we separately validate that Revelio Labs employment and job posting data closely track Office for National Statistics aggregates both over time and across industries. We use two separate Revelio Labs datasets: work profiles and job postings.<sup>2</sup>

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<sup>2</sup>The data used in this paper were downloaded on 12 August 2025.

ket characteristics. We examine heterogeneity along two dimensions: compensation levels (splitting firms and occupations by median pre-treatment salary) and geographic concentration (comparing London-based versus non-London entities). These dimensions are highly policy-relevant, as compensation-level heterogeneity provides a first indication as to whether LLMs may compress or widen wage inequality, while geographic variation tests whether disruption concentrates in London’s knowledge economy or extends throughout the UK labor market.

#### 4.3.1 Heterogeneity by Compensation Level

Table A2 examines how firm-level employment effects vary by average compensation, with high-compensation jobs serving as a proxy for more highly educated white collar work, and low-compensation jobs for relatively low-educated blue collar work. We split firms at the median pre-treatment compensation level. Panel A shows that high-compensation firms experience substantially larger employment reductions than their lower-paying counterparts. A one standard deviation increase in LLM exposure reduces total employment at high-compensation firms by 0.7 percent ( $p = 0.009$ ), driven primarily by a 0.8 percent reduction in low-seniority positions ( $p = 0.005$ ). In contrast, Panel B indicates that low-compensation firms show no significant employment effects, with point estimates near zero for both total employment and low-seniority positions.

This pattern extends to hiring intentions. Table A4 shows that a one standard deviation increase in LLM exposure reduces the probability of posting vacancies by 1.3 percentage points at high-compensation firms ( $p < 0.001$ ), compared to a 0.7 percentage point reduction at low-compensation firms ( $p = 0.001$ ). While both groups curtail hiring, the effect is approximately twice as large at high-compensation firms.

The occupation-level analysis in Table A5 provides further evidence that the employment effects are concentrated among the highly paid. In high-salary occupations, a one standard deviation increase in LLM exposure is associated with a 5.7 percent reduction in job postings ( $p = 0.009$ ) and £782 salary declines ( $p < 0.001$ ), while low-salary occupations show no significant changes in posting volumes and minimal salary effects.

These results suggest that LLM disruption is almost entirely driven by changes

## A1 Additional Tables

Table A1: Effect of LLM exposure on firm-level job listings

|                                  | Job postings         |
|----------------------------------|----------------------|
| LLM exposure x Post              | −0.010***<br>(0.002) |
| Firm fixed-effects               | Yes                  |
| Industry-by-period fixed effects | Yes                  |
| Observations                     | 1,268,013            |
| Adjusted R <sup>2</sup>          | 0.522                |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity. The dependent variable is an indicator equal to one if firm  $i$  posted at least one job vacancy in month  $t$ . LLM exposure is the firm’s standardized pre-treatment exposure score, calculated as the employment-weighted average of occupational LLM exposure scores from Eloundou et al. (2024) using the workforce composition on 30 October 2022. All regressions include firm fixed effects and industry-by-period fixed effects. All other definitions are as in Table 3.

## A2 Heterogeneous Treatment Effects

### A2.1 Firm-Level Employment Stocks

Table A2: Effect of LLM exposure on firm-level job outcomes, high vs. low compensation firms

|                                         | Employees            | Low seniority        | High seniority      | Low seniority (%)   | Av. comp.             |
|-----------------------------------------|----------------------|----------------------|---------------------|---------------------|-----------------------|
| <b>Panel A: High compensation firms</b> |                      |                      |                     |                     |                       |
| LLM exposure x Post                     | −0.007***<br>(0.003) | −0.008***<br>(0.003) | −0.005*<br>(0.003)  | −0.001*<br>(0.0004) | 195.240*<br>(101.538) |
| Firm fixed effects                      | Yes                  | Yes                  | Yes                 | Yes                 | Yes                   |
| Industry-by-period FE                   | Yes                  | Yes                  | Yes                 | Yes                 | Yes                   |
| Observations                            | 634,030              | 633,977              | 634,018             | 633,977             | 634,030               |
| Adj. R-squared                          | 0.975                | 0.973                | 0.974               | 0.963               | 0.956                 |
| <b>Panel B: Low compensation firms</b>  |                      |                      |                     |                     |                       |
| LLM exposure x Post                     | 0.0001<br>(0.003)    | −0.001<br>(0.003)    | −0.00002<br>(0.003) | −0.0004<br>(0.0003) | 69.929<br>(46.938)    |
| Firm fixed effects                      | Yes                  | Yes                  | Yes                 | Yes                 | Yes                   |
| Industry-by-period FE                   | Yes                  | Yes                  | Yes                 | Yes                 | Yes                   |
| Observations                            | 633,983              | 633,978              | 632,313             | 633,978             | 633,983               |
| Adj. R-squared                          | 0.988                | 0.988                | 0.985               | 0.956               | 0.946                 |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for high- and low-compensation firms. Firms are split at the median pre-treatment average compensation level (October 2022). Panel A presents results for firms above the median compensation, Panel B for firms below the median. All other definitions are as in Table 3.

Table A3: Effect of LLM exposure on firm-level job outcomes, London vs. Non-London firms

|                                   | Employees           | Low seniority       | High seniority    | Low seniority (%)    | Av. comp.            |
|-----------------------------------|---------------------|---------------------|-------------------|----------------------|----------------------|
| <b>Panel A: HQ in London</b>      |                     |                     |                   |                      |                      |
| LLM exposure x Post               | -0.001<br>(0.004)   | -0.002<br>(0.004)   | -0.004<br>(0.004) | -0.001<br>(0.001)    | 214.807<br>(132.316) |
| Firm fixed effects                | Yes                 | Yes                 | Yes               | Yes                  | Yes                  |
| Industry-by-period FE             | Yes                 | Yes                 | Yes               | Yes                  | Yes                  |
| Observations                      | 377,175             | 377,129             | 376,920           | 377,129              | 377,175              |
| Adj. R-squared                    | 0.982               | 0.981               | 0.981             | 0.978                | 0.975                |
| <b>Panel B: HQ outside London</b> |                     |                     |                   |                      |                      |
| LLM exposure x Post               | -0.004**<br>(0.002) | -0.005**<br>(0.002) | -0.002<br>(0.003) | -0.001**<br>(0.0003) | 71.962<br>(64.222)   |
| Firm fixed effects                | Yes                 | Yes                 | Yes               | Yes                  | Yes                  |
| Industry-by-period FE             | Yes                 | Yes                 | Yes               | Yes                  | Yes                  |
| Observations                      | 890,838             | 890,826             | 889,411           | 890,826              | 890,838              |
| Adj. R-squared                    | 0.983               | 0.983               | 0.981             | 0.979                | 0.979                |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for firms headquartered in London, and those headquartered elsewhere. Panel A presents results for London-based firms, Panel B for firms based outside London. All other definitions are as in Table 3.

## A2.2 Firm-Level Job Postings

Table A4: Effect of LLM exposure on firm-level job listings, heterogeneity

|                                  | High compensation firms | Low compensation firms | HQ in London         | HQ outside London    |
|----------------------------------|-------------------------|------------------------|----------------------|----------------------|
| LLM exposure x Post              | -0.013***<br>(0.003)    | -0.007***<br>(0.002)   | -0.016***<br>(0.003) | -0.008***<br>(0.002) |
| Firm fixed-effects               | Yes                     | Yes                    | Yes                  | Yes                  |
| Industry-by-period fixed effects | Yes                     | Yes                    | Yes                  | Yes                  |
| Observations                     | 577,489                 | 690,524                | 377,175              | 890,838              |
| Adjusted R <sup>2</sup>          | 0.486                   | 0.554                  | 0.517                | 0.525                |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity across different firm characteristics. Columns 1-2 split firms by median pre-treatment average compensation level. Columns 3-4 split firms by headquarter location (London versus outside London). All other definitions are as in Table A1.

Table A6: Effect of LLM exposure on occupation-level job outcomes, London vs. non-London

|                                    | Job postings (log)  | Salary                   | Salary (log)         | Salary (SD)           |
|------------------------------------|---------------------|--------------------------|----------------------|-----------------------|
| <b>Panel A: Job in London</b>      |                     |                          |                      |                       |
| LLM exposure x Post                | -0.034**<br>(0.017) | -562.818***<br>(139.355) | -0.010***<br>(0.002) | 74.609<br>(132.525)   |
| Occupation fixed-effects           | Yes                 | Yes                      | Yes                  | Yes                   |
| Year-by-month fixed effects        | Yes                 | Yes                      | Yes                  | Yes                   |
| Observations                       | 16,734              | 16,734                   | 16,734               | 16,625                |
| Adjusted R <sup>2</sup>            | 0.954               | 0.922                    | 0.926                | 0.605                 |
| <b>Panel B: Job outside London</b> |                     |                          |                      |                       |
| LLM exposure x Post                | -0.040**<br>(0.018) | -475.947***<br>(141.444) | -0.011***<br>(0.003) | -119.267<br>(111.131) |
| Occupation fixed-effects           | Yes                 | Yes                      | Yes                  | Yes                   |
| Year-by-month fixed effects        | Yes                 | Yes                      | Yes                  | Yes                   |
| Observations                       | 16,812              | 16,812                   | 16,812               | 16,799                |
| Adjusted R <sup>2</sup>            | 0.961               | 0.938                    | 0.945                | 0.683                 |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for London-based jobs and jobs based outside London. Panel A presents results for job openings in London, Panel B for job openings outside London. All other definitions are as in Table 4.

## A2.3 Occupation-Level Job Listings

Table A5: Effect of LLM exposure on occupation-level job outcomes, high vs. low salary occupations

|                                         | Job postings (log)   | Salary                   | Salary (log)         | Salary (SD)              |
|-----------------------------------------|----------------------|--------------------------|----------------------|--------------------------|
| <b>Panel A: High salary occupations</b> |                      |                          |                      |                          |
| LLM exposure x Post                     | -0.057***<br>(0.022) | -782.371***<br>(184.426) | -0.013***<br>(0.003) | -354.844***<br>(131.069) |
| Occupation fixed-effects                | Yes                  | Yes                      | Yes                  | Yes                      |
| Year-by-month fixed effects             | Yes                  | Yes                      | Yes                  | Yes                      |
| Observations                            | 8,413                | 8,413                    | 8,413                | 8,412                    |
| Adjusted R <sup>2</sup>                 | 0.964                | 0.903                    | 0.901                | 0.726                    |
| <b>Panel B: Low salary occupations</b>  |                      |                          |                      |                          |
| LLM exposure x Post                     | -0.020<br>(0.031)    | -112.614<br>(179.137)    | -0.004<br>(0.004)    | 240.391<br>(170.804)     |
| Occupation fixed-effects                | Yes                  | Yes                      | Yes                  | Yes                      |
| Year-by-month fixed effects             | Yes                  | Yes                      | Yes                  | Yes                      |
| Observations                            | 8,402                | 8,402                    | 8,402                | 8,392                    |
| Adjusted R <sup>2</sup>                 | 0.961                | 0.817                    | 0.860                | 0.436                    |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for high- and low-salary occupations. Occupations are split at the median pre-treatment average posted salary (October 2022). Panel A presents results for occupations above the median salary, Panel B for occupations below the median. All other definitions are as in Table 4.